SG No. 128 D_{4h}^{6} $P4/mnc$ [tetragonal]

* plus set: $+[0, 0, 0]$

Table 1: Wyckoff site: `2a`, site symmetry: `4/m..`

No.	position	mapping
1	$[0, 0, 0]$	`[1,2,3,4,9,10,11,12]`
2	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2}\right]$	`[5,6,7,8,13,14,15,16]`

Table 2: Wyckoff site: `2b`, site symmetry: `4/m..`

No.	position	mapping
1	$\left[0, 0, \frac{1}{2}\right]$	`[1,2,3,4,9,10,11,12]`
2	$\left[\frac{1}{2}, \frac{1}{2}, 0\right]$	`[5,6,7,8,13,14,15,16]`

Table 3: Wyckoff site: `4c`, site symmetry: `2/m..`

No.	position	mapping
1	$\left[0, \frac{1}{2}, 0\right]$	`[1,2,9,10]`
2	$\left[\frac{1}{2}, 0, 0\right]$	`[3,4,11,12]`
3	$\left[\frac{1}{2}, 0, \frac{1}{2}\right]$	`[5,6,13,14]`
4	$\left[0, \frac{1}{2}, \frac{1}{2}\right]$	`[7,8,15,16]`

Table 4: Wyckoff site: `4d`, site symmetry: `2.22`

No.	position	mapping
1	$\left[0, \frac{1}{2}, \frac{1}{4}\right]$	`[1,2,7,8]`
2	$\left[\frac{1}{2}, 0, \frac{1}{4}\right]$	`[3,4,5,6]`
3	$\left[0, \frac{1}{2}, \frac{3}{4}\right]$	`[9,10,15,16]`
4	$\left[\frac{1}{2}, 0, \frac{3}{4}\right]$	`[11,12,13,14]`

Table 5: Wyckoff site: `4e`, site symmetry: `4..`

No.	position	mapping
1	$[0, 0, z]$	`[1,2,3,4]`
2	$\left[\frac{1}{2}, \frac{1}{2}, \frac{1}{2} - z\right]$	`[5,6,7,8]`
3	$[0, 0, -z]$	`[9,10,11,12]`
4	$\left[\frac{1}{2}, \frac{1}{2}, z + \frac{1}{2}\right]$	`[13,14,15,16]`

Table 6: Wyckoff site: `8f`, site symmetry: `2..`

No.	position	mapping
1	$\left[0, \frac{1}{2}, z\right]$	`[1,2]`
2	$\left[\frac{1}{2}, 0, z\right]$	`[3,4]`

continued ...

Table 6

No.	position	mapping
3	$\left[\frac{1}{2},\ 0,\ \frac{1}{2}-z\right]$	[5,6]
4	$\left[0,\ \frac{1}{2},\ \frac{1}{2}-z\right]$	[7,8]
5	$\left[0,\ \frac{1}{2},\ -z\right]$	[9,10]
6	$\left[\frac{1}{2},\ 0,\ -z\right]$	[11,12]
7	$\left[\frac{1}{2},\ 0,\ z+\frac{1}{2}\right]$	[13,14]
8	$\left[0,\ \frac{1}{2},\ z+\frac{1}{2}\right]$	[15,16]

Table 7: Wyckoff site: 8g, site symmetry: ..2

No.	position	mapping
1	$\left[x,\ x+\frac{1}{2},\ \frac{1}{4}\right]$	[1,7]
2	$\left[-x,\ \frac{1}{2}-x,\ \frac{1}{4}\right]$	[2,8]
3	$\left[\frac{1}{2}-x,\ x,\ \frac{1}{4}\right]$	[3,5]
4	$\left[x+\frac{1}{2},\ -x,\ \frac{1}{4}\right]$	[4,6]
5	$\left[-x,\ \frac{1}{2}-x,\ \frac{3}{4}\right]$	[9,15]
6	$\left[x,\ x+\frac{1}{2},\ \frac{3}{4}\right]$	[10,16]
7	$\left[x+\frac{1}{2},\ -x,\ \frac{3}{4}\right]$	[11,13]
8	$\left[\frac{1}{2}-x,\ x,\ \frac{3}{4}\right]$	[12,14]

Table 8: Wyckoff site: 8h, site symmetry: m..

No.	position	mapping
1	$[x,\ y,\ 0]$	[1,10]
2	$[-x,\ -y,\ 0]$	[2,9]
3	$[-y,\ x,\ 0]$	[3,12]
4	$[y,\ -x,\ 0]$	[4,11]
5	$\left[\frac{1}{2}-x,\ y+\frac{1}{2},\ \frac{1}{2}\right]$	[5,14]
6	$\left[x+\frac{1}{2},\ \frac{1}{2}-y,\ \frac{1}{2}\right]$	[6,13]
7	$\left[y+\frac{1}{2},\ x+\frac{1}{2},\ \frac{1}{2}\right]$	[7,16]
8	$\left[\frac{1}{2}-y,\ \frac{1}{2}-x,\ \frac{1}{2}\right]$	[8,15]

Table 9: Wyckoff site: 16i, site symmetry: 1

No.	position	mapping
1	$[x,\ y,\ z]$	[1]
2	$[-x,\ -y,\ z]$	[2]
3	$[-y,\ x,\ z]$	[3]
4	$[y,\ -x,\ z]$	[4]
5	$\left[\frac{1}{2}-x,\ y+\frac{1}{2},\ \frac{1}{2}-z\right]$	[5]
6	$\left[x+\frac{1}{2},\ \frac{1}{2}-y,\ \frac{1}{2}-z\right]$	[6]
7	$\left[y+\frac{1}{2},\ x+\frac{1}{2},\ \frac{1}{2}-z\right]$	[7]
8	$\left[\frac{1}{2}-y,\ \frac{1}{2}-x,\ \frac{1}{2}-z\right]$	[8]
9	$[-x,\ -y,\ -z]$	[9]

continued ...

Table 9

No.	position	mapping
10	$[x,\, y,\, -z]$	[10]
11	$[y,\, -x,\, -z]$	[11]
12	$[-y,\, x,\, -z]$	[12]
13	$\left[x+\frac{1}{2},\, \frac{1}{2}-y,\, z+\frac{1}{2}\right]$	[13]
14	$\left[\frac{1}{2}-x,\, y+\frac{1}{2},\, z+\frac{1}{2}\right]$	[14]
15	$\left[\frac{1}{2}-y,\, \frac{1}{2}-x,\, z+\frac{1}{2}\right]$	[15]
16	$\left[y+\frac{1}{2},\, x+\frac{1}{2},\, z+\frac{1}{2}\right]$	[16]